

Test Cases:

Test Case #	Description	Input	Expected Output	Actual Output	Pass/Fail
1	Valid input with a mix of lowercase and uppercase	Max Length: 15 Input String: "Programming"	"Vowel 'a' appears 1 times, first at index 4 Vowel 'o' appears 1 times, first at index 1 Other vowels do not appear in this string"	"Vowel 'a' appears 1 times, first at index 4 Vowel 'o' appears 1 times, first at index 1 Other vowels do not appear in this string"	PASS
2	Input exactly at max length limit	Max Length: 6 Input String: "abcdef"	"Vowel 'a' appears 1 times, first at index 0 Vowel 'e' appears 1 times, first at index 4 Other vowels do not appear in this string"	"Vowel 'a' appears 1 times, first at index 0 Vowel 'e' appears 1 times, first at index 4 Other vowels do not appear in this string"	PASS
3	Input below max length	Max Length: 12 Input String: "hello"	"Vowel 'e' appears 1 times, first at index 1 Other vowels do not appear in this string"	"Vowel 'e' appears 1 times, first at index 1 Other vowels do not appear in this string"	PASS
4	Input with no vowels	Max Length: 8 Input String: "bcd fghjk"	"No vowels appear in this string"	"No vowels appear in this string"	PASS

5	Maximum length input	Max Length: 100 Input String: "i" * 100	Max Length: 100 Input String: "i" * 100	Max Length: 100 Input String: "i" * 100	PASS
6	Input length exceeding specified max length	Max Length: 20 Input String: "abcdefghijklmnopqrst"	"Error: Maximum length exceeded. Please enter a value less than or equal to 20."	"Error: Maximum length exceeded. Please enter a value less than or equal to 20."	PASS
7	Test empty string	Max Length: 10 Input String: ""	"No vowels appear in this string"	"No vowels appear in this string"	PASS